

Study case: EduConnect

LMS Attendance & Leave Management

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Role	Business Analyst
Case type	Academic Case Study
Domain	English Language Center / Education
Focus	Attendance & Leave Management
Primary user	Teacher
Activities	Requirement Analysis · Contextual Observation · Gap Analysis · Solution Design · Workflow & Wireframing

This case was developed as part of the **Software Engineering Specialization** course on Coursera, instructed by **Kenneth Wai-Ting Leung**. The course provided a scenario and direction to propose improvements to an LMS for English language centers.

The given scenario assumed that students would report absences through the LMS. However, observation of an actual English center revealed that students frequently reported absences through Zalo, verbal communication, or counselors instead. This revealed a gap between the **formal system workflow** and **actual operating practices**, potentially leaving attendance data incomplete.

My Contribution:

- **Analyzed** the given requirements and modeled the initial attendance workflow to identify its underlying assumptions.
- **Observed** attendance practices at an English center and compared the observed process with the proposed workflow.
- **Identified** gaps in how absence information was captured and classified.
- **Defined** the problem based on the identified gap.
- **Evaluated** two alternative approaches: LMS-only reporting and teacher-controlled correction, comparing their feasibility against the observed context.
- **Designed** the revised attendance workflow and refined the related requirements.
- **Created** wireframes to demonstrate the proposed solution

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1. Context

1.1. Academic course provided scenario

Directions: Based on real-world observations from English language centers you have attended or studied, propose an innovative and more optimized approach to refine or enhance the system requirements outlined in the problem statement bellow.

In many university-focused English language centers, the management of teaching schedules, student attendance, learning materials, and academic evaluation remains fragmented across multiple platforms, causing communication delays and operational inefficiencies. To address this issue, the proposed solution is the development of "EduConnect LMS – Integrated Academic Management System for English Centers" which connects students, teachers, academic staff, counselors, and administrators onto a unified platform to streamline course delivery and improve academic outcomes.

When students log into the system, they can view their registered courses, track their learning progress, and access official course materials. They can perform online tests, view immediate test results, check their attendance records, and rewatch recorded video lectures for review. Students can also complete homework assignments, submit leave requests, apply for class transfers, issue other administrative requests, and view or edit their personal profile.

Once class sessions begin, teachers access the platform to view their class schedules, class details, and student lists. Teachers can conduct digital attendance, review and approve leave requests, and enter grades for assignments and exams. They are also empowered to open active test modules for students, access supplementary teaching materials, and manage their instructor profile.

To maintain course quality, academic staff manage the core educational assets within the system. They create, upload, and update learning documents and study guides, as well as author, structure, and publish formal assessment tests across various English proficiency levels.

For personalized student care, academic counselors access student information and retrieve contact details to engage in direct, standard communication. Counselors track student performance, offer timely intervention, and provide tailored academic guidance throughout the learning process.

Finally, the administrator is responsible for overseeing overall website and platform configurations. This includes managing user accounts across all roles, regulating access rights, moderating platform content, and tracking system-wide metrics such as student retention rates, assessment trends, system usage statistics, and overall operational efficiency.

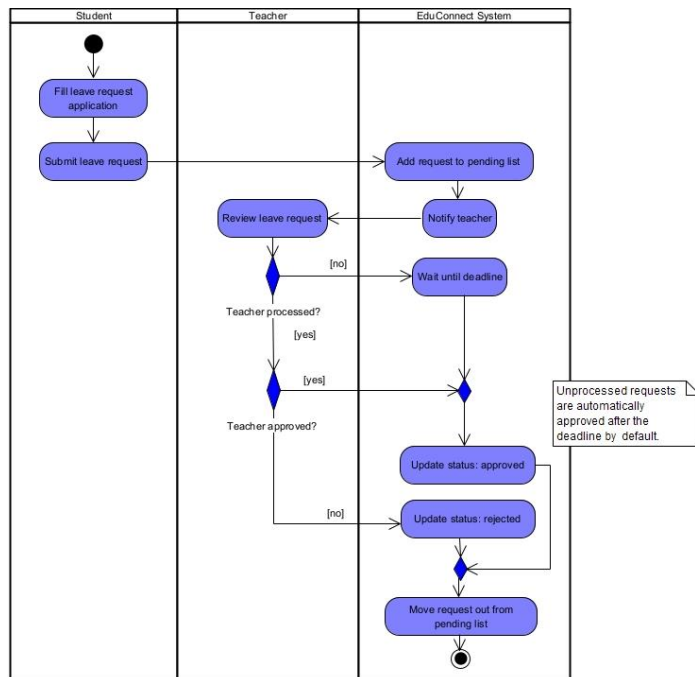
1.2. Relevant requirements extracted from the given scenarios

Case focus: This case focuses specifically on the relationship between **student leave management and attendance**, rather than the entire LMS.

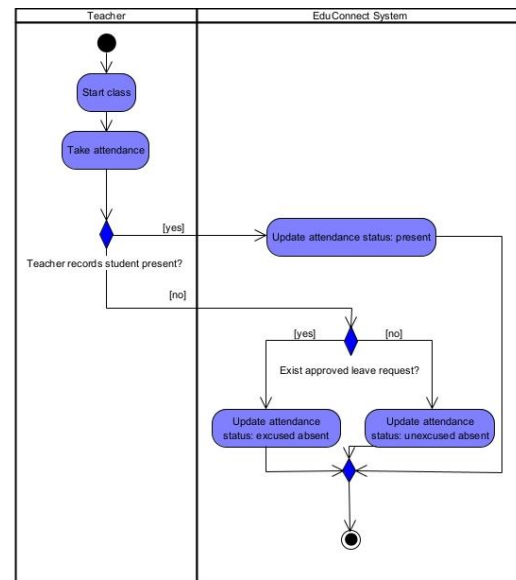
Actor	Functional requirement
Student	Students can submit leave requests through the system.
Teacher	Teachers can review and approve student leave requests.

Activity diagrams for initial workflow:

Leave request & Approve



Attendance



Powered By: Visual Paradigm Community Edition

Initial assumption: The formal leave request is the only channel through which legitimate absence information enters the attendance process.

2. Contextual Observation

To validate whether the proposed leave-management workflow reflected actual student and teacher practices, I conducted contextual inquiries with teachers and a counselor at the English center I attended, complemented by my own observation of student attendance practices.

Observation focus on absence communication, teacher handling, and attendance recording.

2.1. Students' behaviours

- Absence was commonly communicated through **Zalo messages or verbal communication with their teacher or counselor**, rather than a formal system request.
- Formal leave requests through the system were used by some students but were relatively uncommon.
- Students usually provided a reason when reporting an absence.
- For consecutive absences, students could communicate the expected absence period at once.
- Students generally did not wait for formal approval before considering their absence confirmed.
- Some students did not submit any request because they did not perceive a formal leave procedure as necessary.

2.2. Teachers' behaviours

- Teachers could receive absence information through **system requests, Zalo messages, verbal communication, or information relayed by counselors.**
- Teachers still recorded attendance during the class session using the existing management system.
- The formal approval process existed for system-based leave requests.
- In practice, teachers could **use absence information received** outside the system when determining whether an absence should be treated as excused or unexcused.

2.3. Business value

Excused and unexcused absence statuses were operationally relevant because attendance records could affect student attendance-based benefits and the center's handling of attendance-related contractual conditions.

3. Requirement Gap

Initial design assumes	Observed condition
Students submit absence information only through the LMS	Students use multiple channels to report absence
The LMS contains the complete absence information	Teachers may receive relevant information outside the LMS
Attendance decision is based on the system request	Teacher may consider information received through other channels
The formal system requests can be used consistently	Formal system requests are relatively uncommon
Absence information can be centralized through the LMS EduConnect	Centralization depends on students consistently adopting the formal process

The comparison between the initial design and observed conditions shows a clear gap in the absence reporting process. While the system was designed under the assumption that students would report absences solely through LMS EduConnect, observed conditions show students using multiple channels. As a result, teachers receive and consider information outside the system, leaving the LMS incomplete.

4. Problem Definition

The initial workflow may result in incomplete attendance data because absence information received outside the LMS is not captured in the system. This can cause legitimate excused absences to be recorded as unexcused when no formal leave request exists.

Therefore, the system needs a way to maintain accurate attendance records when absence information may be communicated outside the formal LMS workflow.

Problem: How can EduConnect maintain accurate attendance records when students may report absences through channels outside the LMS?

5. Alternative Solutions

5.1. Enforce LMS-only reporting

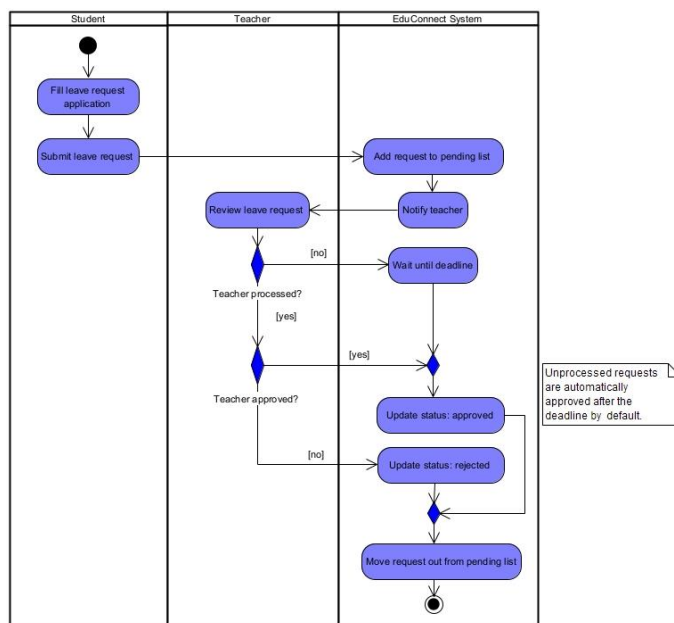
- Concept: Require students to submit all absence information through EduConnect and treat the LMS as the single source of truth.

5.2. Teacher-controlled attendance correction

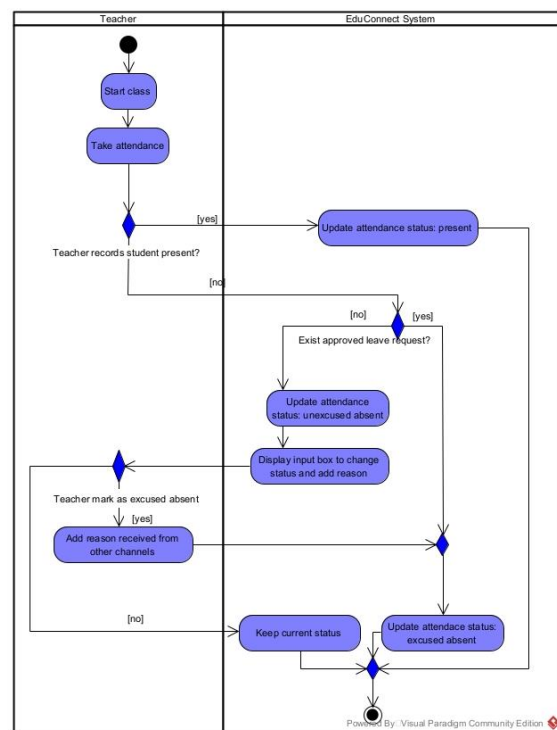
- Concept: Keep the formal LMS leave-request workflow, but allow teachers to correct an absence classification when they have valid information received through other channels.

Activity diagrams for proposed workflow:

Leave request & Approve



Attendance



6. Analyst & Decision

6.1. Solutions advantages and disadvantages

	Enforce LMS-only reporting	Teacher-controlled attendance correction
Advantages	- Centralized data - Consistent workflow - Easier reporting - Clear audit trail - Less manual intervention	- Preserves the formal LMS workflow - Handle real-world absence reporting - Prevents known excused absences from being incorrectly classified - Does not require immediate behavioral change from students
Disadvantages	- Requires consistent student adoption - Requires the center to enforce the policy - Informal channels may continue to be used - Existing user behavior needs to change	- Depends on teacher judgment - Creates a manual operation - Requires a reason/audit trail - Could introduce inconsistency if rules are unclear

The LMS-only approach would provide a cleaner and more centralized workflow, but its effectiveness depends on consistent student adoption of the formal reporting process. This does not fully align with the observed context, where students frequently communicate absences through informal channels.

The teacher-controlled approach therefore retains the formal LMS workflow while providing a controlled mechanism for teachers to incorporate valid absence information received outside the system.

Selected solution: Teacher-controlled attendance correction, with three key rules

1. Formal path remains unchanged: Students can continue submitting formal leave requests through EduConnect.
2. Default attendance behavior: If a student is absent and no approved leave request exists, the system initially records the absence as unexcused.
3. Controlled exception: Teachers can change the status to excused when they have valid absence information from an external channel and must provide a reason for the change.

6.2. Trade-off of selected solution

This approach introduces additional effort for teachers, as they may need to verify fragmented absence information before changing an attendance status. However, under the current operating context, this trade-off provides a practical way to improve attendance data accuracy without assuming that all students will immediately adopt the formal LMS workflow.

The additional effort can be reduced through interface design. For example, the attendance interface can allow teachers to change an absence status and provide a reason within the same workflow, minimizing the need for separate administrative procedures.

6.3. Why not remove the formal leave request workflow?

The solution does not replace the formal leave-request process. Instead, it **extends the existing workflow to cover cases that fall outside the formal path**. Students can continue to use the LMS as the preferred and structured channel for reporting absences. Teachers can also encourage students to gradually adopt the formal process, reducing reliance on external communication over time.

6.4. Long-term consideration

The LMS-only approach may become more appropriate as student adoption of the formal process increases. Therefore, the teacher-controlled mechanism can be viewed not only as an alternative to the LMS-only approach, but also as a **transition mechanism** that allows the system to operate effectively under current conditions while supporting a gradual shift toward centralized absence reporting.

7. Wireframe Design

7.1. Pure wireframe

MA

EduConnect

Faculty Portal

Dashboard

Students

Teachers

Material

Account

Class

Subject

Routine

Attendance

Exam

Notice

Help Center

Logout

Search students, classes...

Attendance Management

Record and track daily student attendance

Today, Oct 24, 2023

Select Class

Select Section

10th Grade

Section A

Mark All Present

P: Present

UA: Unexcused Absent

EA: Excused Absent

Student	Student ID	Attendance Status	Reason
Allotaurus	#00021	☒ P ☐ UA ☐ EA	
Chipmunk	#00022	☒ P ☐ UA ☐ EA	
The Cat	#00023	☐ P ☒ UA ☐ EA	
Nightingale	#00024	☒ P ☐ UA ☐ EA	
Jaguar	#00025	☐ P ☐ UA ☒ EA	Medical Appointment
Eagle	#00026	☐ P ☐ UA ☒ EA	Reason is required for Excused Absent

Last save: Oct 24, 2023, 08:30 AM by Teacher Da Bird

Save Attendance

7.2. Wireframe with callout

MA

EduConnect

Faculty Portal

Search students, classes...

Dashboard

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Eagle	#00026	☐ P ☐ UA ☒ EA	Reason is required for Excused Absent

Last save: Oct 24, 2023, 08:30 AM by Teacher Da Bird

Save Attendance

Callout 1 - Bulk action

Teachers can mark the whole class as Present and only change students who are absent.

Callout 2 - Controlled correction

Teachers can decide whether an absence is excused or unexcused when they have valid absence information.

Callout 3 - Automatic pre-fill

If an approved leave request exists, the system automatically marks the student as Excused and pre-fills the reason.

Callout 4 - Reason required

A reason is required when changing an absence to Excused, providing context for the manual correction.

8. Expected Impact

- Absence information received outside the LMS is incorporated directly into the formal attendance record.
- Reduces incorrectly recorded unexcused absences when students miss formal leave request submissions.
- Approved leave requests automatically update attendance records and pre-fill reasons.
- Teachers retain the ability to adjust statuses for exceptional cases while logging a reason.
- Keeps existing LMS workflows active, allowing the center to transition students toward centralized reporting without compromising record accuracy today.

9. Reflection

First, I learned that collected requirements should not always be treated as an accurate representation of how a process actually works. Requirements may contain implicit assumptions that appear reasonable at first but fail to cover real-world exceptions. In this case, I initially assumed that the formal LMS leave-request workflow would capture the necessary absence information. Observing how students and teachers actually communicated absences revealed cases that the initial requirement did not account for. This reinforced the importance of validating requirements against actual user practices rather than relying solely on documented requirements.

Second, I learned to consider **auditability** as part of solution design. My initial design focused on allowing teachers to manage attendance but did not sufficiently consider how manual changes should be explained or traced. Introducing a required reason for attendance corrections made me recognize that whenever a system gives users the ability to override or modify important records, it should also provide an appropriate level of accountability.

Finally, I learned that **similar systems cannot automatically be treated as reliable references for another context**. Different educational organizations may have significantly different operating practices, even when they provide similar services. A workflow that works well in a university may be appropriate because attendance procedures are formally enforced, while the same assumption may not hold in an English center. Therefore, when designing a system for a specific organization, I should validate its actual operating context instead of assuming that practices from related systems or other organizations apply unchanged.

10. Limitation

While this approach addresses key attendance gaps, it introduces several trade-offs. Teachers may experience an increased workload, as they are now responsible for manually verifying and recording absence information received through external channels. Furthermore, the solution does not eliminate external communication tools like Zalo or verbal notices, it simply offers a pathway to pull that data into the formal system.